CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 17: Circle

Topic-wise short notes with solved examples

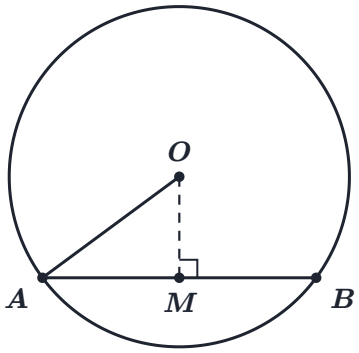
The parts of a circle, the properties of chords and their distances from the centre, and the relation between equal chords, equal arcs and the angles they subtend at the centre.

1 Terms

- A **circle** is the set of all points in a plane at a fixed distance (the **radius**) from a fixed point (the **centre**).
- **Chord**: a line segment joining two points of the circle. **Diameter**: a chord through the centre; it is the longest chord and equals twice the radius.
- **Arc**: a part of the circle. A chord divides the circle into a **minor arc** and a **major arc**.
- **Segment**: the region between a chord and its arc. **Sector**: the region between two radii and an arc.
- **Concentric circles** have the same centre. **Congruent (equal) circles** have equal radii.
- **Circumference** $= 2\pi r$.

2 Chord and perpendicular from the centre

- **Theorem 1**: the perpendicular from the centre to a chord **bisects the chord**.
- **Theorem 2 (converse)**: the line joining the centre to the mid-point of a chord is **perpendicular to the chord**.
- The perpendicular bisector of any chord passes through the centre.
- **One and only one circle** passes through three points that are not in a straight line. Its centre is where the perpendicular bisectors of the segments joining the points meet.



OM is perpendicular to chord AB, so AM = MB

- If the radius is r, the chord has length c and its distance from the centre is d:

$$r^2 = d^2 + \left(\frac{c}{2}\right)^2$$

Example 2.1 A chord of length 24 cm is drawn in a circle of radius 13 cm. Find its distance from the centre.

SOLUTION

$d^2 = 13^2 - 12^2 = 25$

5 cm

Example 2.2 A chord is 6 cm from the centre of a circle of radius 10 cm. Find its length.

SOLUTION

Half the chord $= \sqrt{100 - 36} = 8$ cm

16 cm

Example 2.3 A chord of length 16 cm is 15 cm from the centre. Find the radius.

SOLUTION

$r^2 = 15^2 + 8^2 = 289$

17 cm

3 Equal chords

- **Theorem 3**: equal chords of a circle (or of equal circles) are **equidistant from the centre**.
- **Theorem 4 (converse)**: chords that are equidistant from the centre are **equal**.
- Of two chords, the one **nearer** to the centre is **longer**. The diameter is the longest chord.

Example 3.1 Two chords of a circle are each 10 cm long. One is 12 cm from the centre. How far is the other?

SOLUTION

12 cm

4 Two parallel chords

- Draw the perpendicular from the centre to both chords. It bisects each.
- If the chords are on **opposite sides** of the centre, the distance between them is the **sum** of their distances from the centre. If they are on the **same side**, it is the **difference**.

Example 4.1 In a circle of radius 13 cm, two parallel chords have lengths 10 cm and 24 cm. Find the distance between them when they are (i) on opposite sides of the centre (ii) on the same side.

SOLUTION

Distances from the centre: $\sqrt{169 - 25} = 12$ cm and $\sqrt{169 - 144} = 5$ cm.

(i) 17 cm (ii) 7 cm

Example 4.2 Two parallel chords of lengths 30 cm and 16 cm are on opposite sides of the centre and 23 cm apart. Find the radius.

SOLUTION

Let the distances from the centre be x and $23 - x$. Then $x^2 + 15^2 = (23 - x)^2 + 8^2$.

$225 = 529 - 46x + 64$, so $46x = 368$ and $x = 8$.

$r = \sqrt{64 + 225} = 17$ **cm**

5 Two intersecting circles

- The line joining the centres of two intersecting circles is the **perpendicular bisector of their common chord**.

Example 5.1 Two circles of radii 10 cm and 17 cm intersect, and their centres are 21 cm apart. Find the length of the common chord.

SOLUTION

Let the common chord be at distance x from the first centre: $100 - x^2 = 289 - (21 - x)^2$.

$100 - x^2 = 289 - 441 + 42x - x^2$, so $42x = 252$ and $x = 6$.

Half the chord $= \sqrt{100 - 36} = 8$ cm

16 cm

6 Arcs and chords

- In the same circle or in equal circles:

Given	Result
Equal chords	They subtend equal angles at the centre
Chords subtending equal angles at the centre	The chords are equal
Equal arcs	They subtend equal angles at the centre
Equal arcs	Their chords are equal
Equal chords	They cut off equal arcs

- The angle subtended at the centre is proportional to the length of the arc.

Example 6.1 AB is a side of a regular hexagon and AC is a side of a regular octagon, both inscribed in the same circle with centre O. Find $\angle AOB$ and $\angle AOC$.

SOLUTION

The sides of a regular polygon are equal chords, so they subtend equal angles at the centre.

$\angle AOB = \frac{360^\circ}{6} = 60^\circ$ **and** $\angle AOC = \frac{360^\circ}{8} = 45^\circ$

Example 6.2 In a circle with centre O, chord AB equals the radius. Find $\angle AOB$.

SOLUTION

$OA = OB = AB$, so triangle OAB is equilateral.

60°

Example 6.3 AB and CD are equal chords of a circle with centre O. If $\angle AOB = 80^\circ$, find $\angle OCD$.

SOLUTION

$\angle COD = 80^\circ$ and $OC = OD$, so $\angle OCD = \dfrac{180^\circ - 80^\circ}{2}$.

50°

7 Common mistakes

- Using the whole chord instead of **half** the chord in the right triangle.
- Adding the distances for parallel chords on the same side of the centre (subtract), or subtracting for opposite sides (add).
- Saying the longer chord is farther from the centre. It is nearer.
- Forgetting that the perpendicular must come from the **centre** to bisect the chord.
- Confusing a segment with a sector.

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